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Sending The Wrong Price Signal: Why Do Some Brand-Name Drugs Cost Medicare Beneficiaries Less Than Generics?

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ABSTRACT The current Medicare Part D benefit may require greater out-of-pocket spending for beneficiaries filling prescriptions for higher-price generic drugs, compared to those filling brand-name counterparts. This can occur among patients who reach the catastrophic coverage phase under the Part D benefit, when differences between the prices for generic and brand-name drugs are not large. This scenario may be common with specialty drugs (typically high-price products used to treat rare or complex conditions), when the number of generic manufacturers or clinical alternatives are limited. In this study we demonstrated that patients would pay more out of pocket for generic medications than for brand-name drugs in these cases, driven by manufacturer discounts provided in the Medicare Part D coverage gap. Overpayments for specialty generic drugs relative to brand-name drugs ranged from \$869 to \$1,072 in 2019, despite lower point-of-sale prices for these drugs. Policy makers should consider modifying the Part D benefit to increase incentives for generic drug use.

Generic competition has historically been a successful strategy for lowering prescription drug expenditures, with US savings from generics exceeding \$265 billion in 2017.¹ However, to fully realize savings and reduce out-of-pocket spending for beneficiaries, generics (typically small-molecule drugs) and biosimilars (follow-on or generic versions of large-molecule biologics) need to be offered at lower net prices to payers than their brand-name counterparts are, to gain preferred placement on formularies. This may be particularly challenging for small-molecule specialty drugs (high-price drugs used to treat rare or complex diseases) and biosimilars, which have less robust competition than traditional small-molecule drugs do.^{2–5}

For Medicare beneficiaries needing small-molecule specialty drugs or biologics, price differences between generics or biosimilars and

their brand-name counterparts may be relatively modest, compared with traditional generic drugs. Furthermore, since 2012, as part of the Affordable Care Act's effort to close the Part D coverage gap (the phase of the Medicare prescription drug benefit where patients were required to pay 100 percent of the cost of their medications), patients who use brand-name drugs have received a manufacturer discount that is counted toward their out-of-pocket spending. This allows beneficiaries who use brand-name drugs to reach the catastrophic coverage phase (where patients pay 5 percent of the drug's point-of-sale price through the end of the calendar year) with lower actual out-of-pocket spending. For example, in 2019 users of brand-name drugs who entered the coverage gap and received the manufacturer discount would reach the catastrophic coverage phase after spending only \$982 out of pocket versus \$3,730 spending for users of generic drugs, who would receive no

such discount.

As a result of this concern, in early 2018 the Bipartisan Budget Act (BBA) modified the Part D benefit to ensure that beneficiaries would not pay more for biosimilars than for their brand-name counterparts. While this change corrected the misaligned incentives for biosimilars, it did not extend the protection to generic drugs. In this article we describe how the current Medicare Part D benefit design can lead to higher out-of-pocket spending for some generic drugs relative to their brand-name counterparts. As part of our study, we estimated how the BBA changed out-of-pocket spending for users of brand-name and generic or biosimilar drugs.

Study Data And Methods

We used data for the first quarter of 2018 from the Medicare Formulary Files to compare prices, formulary coverage, and expected annual out-of-pocket spending for stand-alone and Medicare Advantage Part D enrollees using pairs of brand-name and generic or biosimilar drugs, assuming no other medication use. We used a convenience sample of nine brand-name products with generics or biosimilars approved in 2015–16 that represent a range of approval pathways, potential market sizes, and point-of-sale prices (exhibit 1 and online appendix exhibit A1).⁶ For each of these products, we calculated out-of-pocket spending using spending thresholds from the 2019 standard benefit design.⁷ For brand-name and generic drugs, we assumed that patients paid a \$415 deductible, 25 percent coin-

surance in the initial phase and coverage gap, and 5 percent coinsurance in the catastrophic phase. Although the 2019 standard benefit applies 37 percent coinsurance for generic drugs purchased in the coverage gap, this drops to 25 percent in 2020 after the coverage gap is closed. Therefore, to provide a more conservative estimate of out-of-pocket spending differences between brand-name and generic products, we elected to use the lower cost-sharing percentage for generic products. In 2019 patients remain in the initial coverage phase until drug spending exceeds \$3,820 and in the coverage gap until their cumulative out-of-pocket spending reaches \$5,100. Before the BBA, users of brand-name drugs received a 50 percent discount in the coverage gap that was applied to out-of-pocket spending to reach catastrophic coverage, while users of biosimilars and generic drugs did not receive such a discount. After the BBA, this discount increased from 50 percent to 70 percent and applied to both brand-name drugs and biosimilars but not to other generic drug products.

In addition to comparing product-specific out-of-pocket spending, we also addressed two broader questions. First, what level of discount is needed from the brand-name product price to make generic drugs less costly than brand-name drugs for high-cost enrollees? Second, at what level of drug spending do beneficiaries have financial incentives to switch to brand-name drugs? For the former, we estimated differences in out-of-pocket spending on brand-name versus generic drugs assuming discounts of 10–90 per-

EXHIBIT 1

Characteristics of brand-name and generic or biosimilar products in the study

Brand-name drug	Generic or biosimilar	Drug category	Median point-of-sale price (\$)		Generic or biosimilar savings (% discount from brand-name price)	Part D plans covering the drug (%)	
			Brand-name drug	Generic or biosimilar		Brand-name drug	Generic or biosimilar
Crestor	Rosuvastatin	Traditional	267	16	94.0	18.1	97.6
Lantus	Basaglar	Traditional	383	350	8.6	96.8	16.8
Abilify	Aripiprazole	Traditional	908	377	58.5	2.9	100.0
Invega	Paliperidone	Traditional	1,130	459	59.4	19.4	97.8
Remicade	Inflectra	Biosimilar	1,175	967	17.7	96.3	21.2
Neupogen	Zarxio	Biosimilar	3,277	2,807	14.3	88.5	58.6
Copaxone	Glatiramer	Specialty	7,152	4,870	31.9	68.7	59.5
Nilandron	Nilutamide	Specialty	7,665	5,045	34.2	18.6	99.2
Gleevec	Imatinib	Specialty	10,604	8,054	24.0	19.1	99.1

SOURCE Authors' analysis of Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files for the first quarter of 2018. **NOTES** Traditional generics are small-molecule drugs that are chemically equivalent to the brand-name version. Biosimilars are "generic" versions of biologic products. Specialty generics are high-price products used to treat rare or complex diseases. Because biosimilars have unique naming conventions for the nonproprietary drug name, we use the proprietary name for the biosimilars studied (shown as capitalized in text). The median point-of-sale price is aggregated across all Part D plans available in the study quarter for a standard fill at the default dose recorded in the Part D plan finder. Plans may elect to cover the brand-name drug only, the generic or biosimilar only, or both products, so the percentages in the last two columns might not sum to 100. Appendix exhibit A1 shows additional characteristics (see note 6 in text).

cent for enrollees with annual drug expenditures of \$27,052.⁴ This value represents the average annual Part D spending for the one million enrollees who are estimated to be high cost and who did not receive low-income subsidies in Part D in 2015. For the latter evaluation, we assumed a fixed generic drug discount of 61 percent off brand-name drug prices, which corresponds to an expected price decrease with up to four manufacturers.^{3,8} We then plotted the differences in out-of-pocket spending on brand-name versus generic drugs as total annual spending on drugs increased from \$3,000 to \$120,000 per year.

SENSITIVITY ANALYSES In primary analyses we estimated out-of-pocket spending assuming that users of generic drugs paid only 25 percent coinsurance in the coverage gap (instead of the 37 percent required under the standard benefit). This approach provided a more conservative estimate and reflected the lower coinsurance for generics expected in 2020 and beyond. In a sensitivity analysis we recalculated out-of-pocket spending differences using 37 percent coinsurance in the coverage gap and found that the results were similar (data not shown).

In scenario analyses we used a fixed discount (61 percent)⁸ and varied drug spending to determine when the use of generic drugs resulted in out-of-pocket savings for beneficiaries. Because the discount may be higher or lower, depending on the level of competition within the drug class, we conducted two sensitivity analyses, assuming that prices were reduced by only 31 percent (the average reduction observed among specialty drugs in the current study) or by 80 percent (a generous assumption of price differences within the postapproval time frame studied).⁹ (See appendix exhibits A2 and A3 for estimates of the range of drug spending at which generics provide cost savings over brand-name drugs for the specified discount levels.)⁶

LIMITATIONS Our study had several limitations. First, for the analysis of individual product price changes, we selected a convenience sample of brand-name and generic or biosimilar products with recent Food and Drug Administration (FDA) approval, which might not fully represent trends across all products. The prices used represent average point-of-sale prices and do not include rebates or other discounts received by the plan after the point of sale.¹⁰ Because our focus is on patients' out-of-pocket spending, we believe that point-of-sale prices are more relevant in this setting.

Second, we also lacked information on patients' use of financial assistance programs, which could have lowered out-of-pocket spending.

Third, to simplify the comparison, for product-

Policy makers should modify the Part D benefit to ensure that patients who use generic drugs are not penalized for doing so.

specific estimates we assumed that patients received twelve monthly fills using the default dose obtained in the Part D plan finder and used only one product for the year. These decisions might not reflect the clinical use of products and may overestimate patients' payments for an individual drug in some situations. However, our primary focus was on relative price changes across brand-name drugs and their generic or biosimilar counterparts over time, which minimized the impact of these limitations on our broader findings. Furthermore, we demonstrate how the level of drug spending and assumptions regarding price differences for generic drugs could lead to similar results for patients using a variety of brand-name or generic products.

Fourth, we did not account for other therapeutic options available to patients, which could have driven coverage for some products.

Fifth, we used the Medicare Part D standard benefit design for estimating out-of-pocket spending, a design that employs 25 percent coinsurance in the initial coverage phase of the benefit. Plans could instead use flat fees (copayments), which might have lowered out-of-pocket spending in this phase of the benefit. However, we found that fewer than half of the plans for each specialty or biosimilar product used copayments, which suggests that this would not materially affect our estimates.

Study Results

Median point-of-sale prices for a standard fill for brand-name drugs were consistently higher than those for generics and biosimilars, with savings ranging from 94.0 percent for Crestor/rosuvastatin to 8.6 percent for Lantus/basaglar (exhibit 1). We calculated that the average point-of-sale price for generics and biosimilars was 38.1 percent less than the brand-name drug price across all products studied (data not shown).

In the period before the BBA, traditional ge-

nerics reduced average annual out-of-pocket spending relative to brand-name drugs by \$100 for basaglar/Lantus and as much as \$1,219 for aripiprazole/Abilify (exhibit 2). We calculated that the average reduction in annual out-of-pocket spending for the four traditional generics in our study was \$709 (see appendix exhibit A4).⁶ However, during this same time period (pre-BBA), biosimilars and specialty generic drugs required higher patient out-of-pocket spending relative to that for brand-name drugs. This increase ranged from \$591 for imatinib/Gleevec to \$1,949 for Zarxio/Neupogen (exhibit 2). For example, patients filling a year of the biosimilar Inflectra would spend \$4,097 out of pocket versus \$2,858 for the brand-name drug Remicade (see appendix exhibit A4 for estimated annual spending by product).⁶

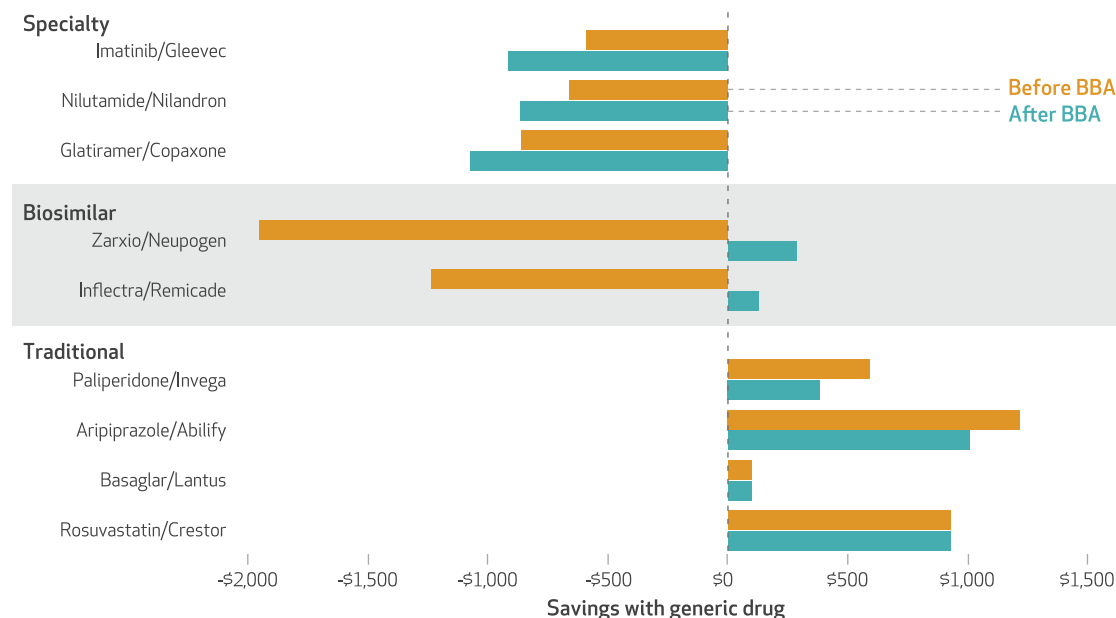
After the BBA, starting in early 2019, annual out-of-pocket spending for patients using biosimilars is decreasing. Patients using the biosimilar Inflectra, for example, will save \$1,573 in annual out-of-pocket spending on this drug relative to their pre-BBA spending (with \$2,524 in out-of-pocket spending after this policy change),

assuming that prices remain similar to those observed in 2018 (appendix exhibit A4).⁶ However, out-of-pocket spending for generics is increasing relative to that for brand-name drugs post BBA because of manufacturer discounts on brand-name drugs increasing from 50 percent to 70 percent. For users of brand-name drugs, this results in additional annual savings of approximately \$209. In 2019, overpayments for specialty generic drugs relative to brand-name drugs ranged from \$869 for nilutamide/Nilandron to \$1,072 for glatiramer/Copaxone.

In scenario analyses, assuming that patients have \$27,052 in prescription drug spending during the calendar year⁴ under the standard Part D benefit post-BBA, we found that discounts for nonbiosimilar generic drugs would need to be at least 65 percent of the brand-name drug price to ensure that patients would have lower annual out-of-pocket spending when using generics (exhibit 3). Similarly, if we assumed a discount of 61 percent (the expected level of discounting for products with four generic or biosimilar manufacturers),⁸ we found that patients who had annual prescription drug spending between

EXHIBIT 2

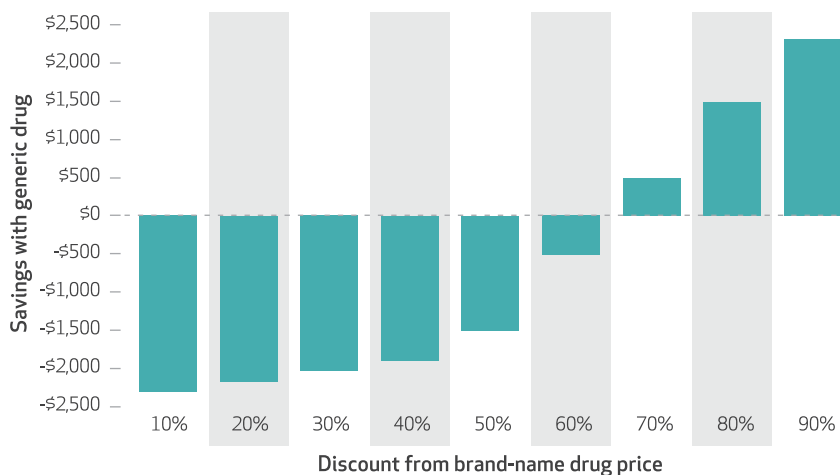
Annual out-of-pocket spending differences for generic and biosimilar drugs and brand-name equivalents covered by Medicare Part D before and after passage of the Bipartisan Budget Act (BBA) of 2018



SOURCE Authors' analysis of Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files for the first quarter of 2018. **NOTES** Negative dollar amounts represent higher out-of-pocket spending on generic and biosimilar drugs relative to brand-name equivalents, while positive amounts represent lower out-of-pocket spending. Traditional generics, biosimilars, and specialty generics are explained in the notes to exhibit 1. In the pairs of drug names, the second name is that of the brand-name drug. Out-of-pocket spending estimates are based on 2018 point-of-sale prices. We assumed that patients paid a \$415 deductible, 25 percent coinsurance in the initial phase and coverage gap, and 5 percent coinsurance in the catastrophic phase (the periods are defined in the text). Before the BBA, a 50 percent discount was applied to out-of-pocket spending for brand-name drugs in the coverage gap. After the BBA, this discount increased to 70 percent and also applied to biosimilars. Appendix exhibit A4 shows annual out-of-pocket spending by product (see note 6 in text).

EXHIBIT 3

Discounts from brand-name drug prices needed to achieve savings on annual out-of-pocket spending on generic drugs for high-cost Medicare beneficiaries with Medicare Part D



SOURCE Authors' calculations of out-of-pocket spending under Medicare Part D, using the standard benefit in 2019. **NOTES** Negative dollar amounts represent higher out-of-pocket spending on generic drugs relative to brand-name equivalents, while positive amounts represent lower out-of-pocket spending. Authors' calculations of out-of-pocket spending for brand-name and generic drugs are based on the assumptions in the notes to exhibit 2. We assumed annual drug spending of \$27,052, which corresponds to the level of spending by high-cost Medicare enrollees without low-income subsidies who were in the Part D program in 2015 (see Medicare Payment Advisory Commission. Report to the Congress [note 4 in text]).

approximately \$22,000 and \$80,000 would save money by using brand-name drugs instead of nonbiosimilar generic drugs (exhibit 4). Considering the per fill expense of many specialty drugs, this level of spending would be likely among people with complex illnesses such as cancer, rheumatoid arthritis, or multiple sclerosis.

Discussion

In this study we found that patients using high-price specialty generic drugs would spend more out of pocket for these products under the current Medicare Part D benefit than they would for the drugs' brand-name counterparts, both before and after the provisions of the Bipartisan Budget Act of 2018 were in place. While the BBA lowered patients' out-of-pocket spending for biosimilars, it also increased discounts on brand-name products and did nothing to reduce out-of-pocket spending for nonbiosimilar generic drugs. In some cases, this resulted in higher out-of-pocket spending for generics than for brand-name drugs, a dynamic driven by higher point-of-sale prices and smaller price differentials between brand-name and generic specialty drugs, compared to traditional generic entrants. In these cases, brand-name manufacturer discounts provided in the coverage gap reduce annual out-of-pocket spending for beneficiaries by

moving users of brand-name drugs more quickly into the catastrophic phase of coverage. Notably, anticipated increases in patients' out-of-pocket spending requirements for entering catastrophic coverage—from \$5,100 in 2019 to \$6,350 in 2020—will further exacerbate this issue.¹¹

There are several reasons to be concerned about this issue even if plans were to achieve similar net prices (after rebates and discounts) for brand-name drugs and their generics or biosimilars. First, incentives for the use of brand-name specialty products may have important market consequences if they result in decreased market share for generics or discourage additional generic entrants, further impeding price decreases.³ Given the lack of marketing and promotional spending on generic and biosimilar products and the lower levels of competition generally observed for products used for rare diseases,³ this is an important risk. Second, patients might not have the option in their Part D plans to switch between brand-name and generic drugs to maximize out-of-pocket savings, because plans likely offer only one of these products and not both. Finally, patients in states with generic substitution laws might have difficulty obtaining brand-name drugs even if these are available on their plans for lower out-of-pocket prices unless their providers include “dispense as written” on prescriptions. Encouraging providers to use “dispense as written” in some cases would be ill advised, given the importance of promoting generic drug use more broadly for containing prescription drug spending.

We identify two potential policy solutions. Eliminating manufacturer discounts from out-of-pocket spending calculations would reduce barriers to generic drug use by leveling the playing field between brand-name and generic out-of-pocket spending. However, eliminating discounts would increase out-of-pocket spending for patients using brand-name drugs, especially when no generic competitors exist. Related recommendations to eliminate manufacturer discounts from out-of-pocket spending calculations have been made by the Medicare Payment Advisory Commission⁴ and were included in the recent drug pricing blueprint of the administration of President Donald Trump¹² and the president's 2019 budget.¹³ In these proposals, adjustments to the coverage gap discount are combined with a cap on patients' out-of-pocket spending and a revision to the benefit that requires plans to pay a higher proportion of drug costs in the catastrophic coverage phase. Such efforts could provide important financial relief to patients with very high out-of-pocket spending but need to be implemented cautiously, given the lack of price pressure that would result from capping

out-of-pocket spending for beneficiaries.

Alternatively, the Part D benefit could extend the discounts for brand-name drugs and biosimilars filled in the coverage gap to generics. However, the question of who would finance such discounts should be carefully considered. If generic drug companies participated in the coverage gap discount program, it would likely increase prices for these products. Alternatively, if payers were asked to contribute, it might further disadvantage generic products for formulary inclusion and placement.

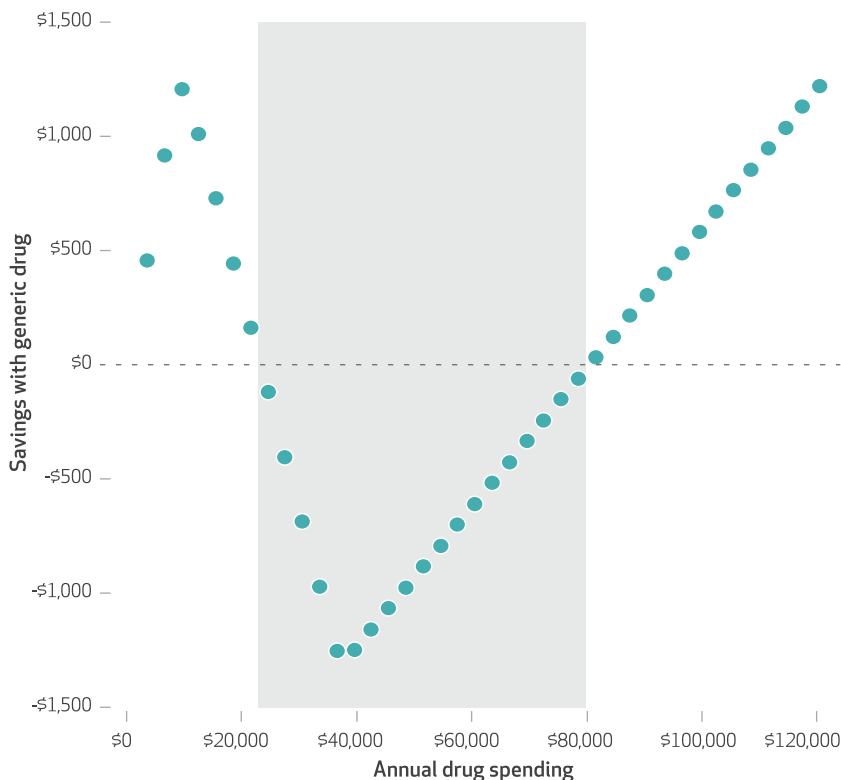
To date, guidance from the Centers for Medicare and Medicaid Services (CMS) on biosimilar coverage under Part D has not comprehensively addressed variations in the FDA approval processes for biosimilar products in the US. In a 2015 memorandum to Part D sponsors, CMS indicated that Part D formulary changes involving biosimilar products would be subject to review on an individual basis based on the FDA-approved label.¹⁴ This guidance suggested that in general, the FDA designation for biosimilar products would inform coverage decisions. Notably, products listed in the FDA's Purple Book (products approved under Section 351[k] of the Public Health Service Act, or the abbreviated licensure pathway) would be categorized as biosimilars under the Medicare Part D benefit.^{14,15} Of the products included in our analysis, only Zarxio and Inflectra are currently approved as biosimilars in both the US and Europe. Basaglar, in contrast, is not considered a biosimilar product under the Medicare Part D benefit in this guidance as it was not approved or licensed under the abbreviated licensure pathway.¹⁶ Considering the distinction between biosimilars and generics within the Part D benefit, where the former is now subject to a substantial discount and the latter is not, it will be important to ensure that approval pathways do not create artificial barriers to the adoption of lower-price generic or biosimilar medications.

Conclusion

Manufacturer discounts in the Medicare Part D coverage gap currently create a perverse incentive for patients to select higher-price brand-name drugs instead of their generic or biosimilar counterparts in some cases. The Bipartisan Budget Act of 2018 corrected these incentives for biosimilar products but will exacerbate out-of-pocket spending differences between special-

EXHIBIT 4

Difference in out-of-pocket spending with generic drug use relative to brand-name drug use, by total annual drug spending for beneficiaries in Medicare Part D



SOURCE Authors' calculations of out-of-pocket spending across varying levels of assumed annual drug spending under Medicare Part D, using the standard benefit in 2019. **NOTES** Negative dollar amounts represent higher out-of-pocket spending on generic drugs relative to brand-name equivalents, while positive amounts represent lower out-of-pocket spending. We assumed that generic drugs received a 61 percent discount from brand-name drugs. Authors' calculations of out-of-pocket spending for brand-name and generic drugs are based on the assumptions in the notes to exhibit 2. The negative amounts represent savings when filling prescriptions for brand-name drugs. Each data point in the figure represents the difference in out-of-pocket spending for filling generic relative to brand-name drugs as total annual drug spending increases from \$3,000 to \$120,000. The shaded area represents annual spending levels where out-of-pocket spending on generic drugs is higher than it would be on the brand-name counterpart.

ty brand-name and generic medications. Combined with pending increases in beneficiary spending requirements in the coverage gap and no out-of-pocket spending limit in Part D, higher-price generic drugs are likely to be unaffordable to beneficiaries. Policy makers should modify the Part D benefit to ensure that patients who use generic drugs are not penalized for doing so, whether that be through removing the manufacturer discount from out-of-pocket spending calculations or extending the discount to all generic drugs. ■

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NOTES

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